INCH-POUND

MIL-T-27/221B
20 April 1993
SUPERSEDING
MIL-T-27/221A
7 October 1987

MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, TF5T03ZZ

(B) Inactive for new design after the date of this specification

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-T-27

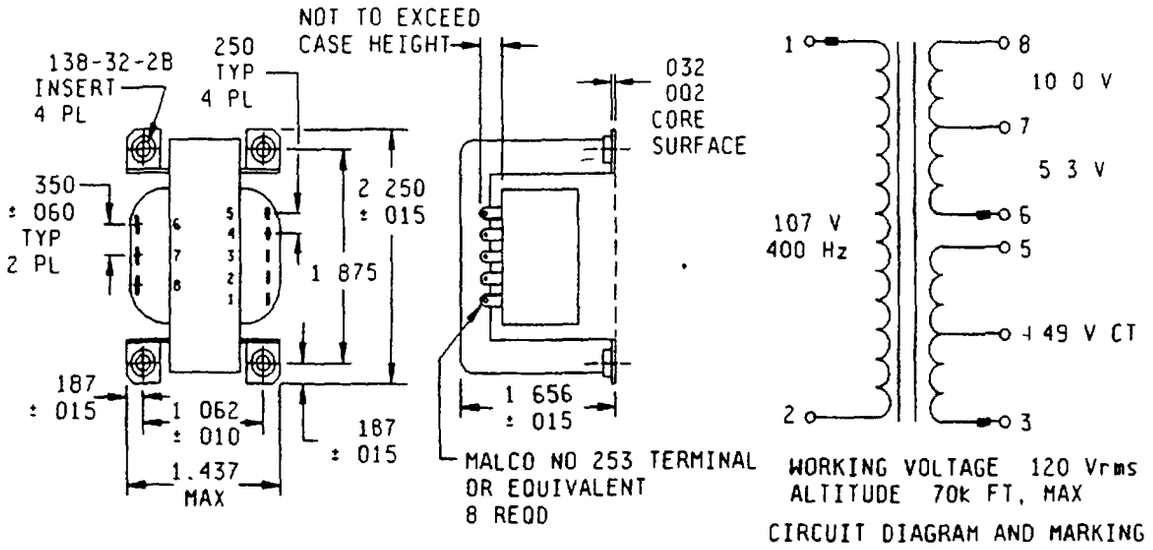


FIGURE 1 Dimensions and configurations

(B) denotes changes

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Inches	mm	Inches	mm
002	0 05	250	6 35
010	0 25	350	8 89
015	0 38	1 062	26 97
032	0 81	1 437	36 50
060	1.52	1 656	42 06
138	3 51	1 875	47 62
187	4 75	2 250	57 15

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for information only
3 Marking shall be on the sides of the case

FIGURE 1 Dimensions and configurations - Continued

REQUIREMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the windings)

Electrical ratings

Primary voltage (1-2) 107 volts, 400 Hz ±20

Primary current (1-2) 0 35 ampere

Voltamperes 37 voltamperes

Secondary voltage

(3-5) 49 volts ct
(6-7) 5 3 volts
(7-8) 10 0 volts

Secondary current

(3-5) 0 010 ampere
(6-7) 1.63 amperes
(7-8) 0 60 ampere

Working voltage 120 volts rms

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Encapsulated

Terminals Solder terminal, Malco no 253 or equivalent

Terminal height Not to exceed case height

Weight 7 5 ounces maximum

Altitude 70,000 feet maximum

Operating temperature range -55°C to +170°C

Terminal strength MIL-STD-202, method 211, test condition A, 5 pounds

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Dielectric withstanding voltage

At sea level 1,000 volts rms
At barometric pressure 300 volts rms

Electrical characteristics

No load With 107 volts rms, 400 Hz across (1-2), the current in (1-2) shall not exceed 150 milliamperes and the power in (1-2) shall not exceed 18 watts

Voltage across

(3-5) 49 volts ±10 percent
(6-7) 5 3 volts ±10 percent
(7-8) 10 0 volts ±10 percent

Rated load With 107 volts rms, 400 Hz across (1-2), the voltage across

(3-5) 49 volts ct ±10 percent
(6-7) 5 3 volts ±10 percent
(7-8) 10 0 volts ±10 percent

DC resistance

(1-2) 7 5 ohms, maximum
(3-5) 2 5 ohms, maximum
(6-7) 0 186 ohm, maximum
(7-8) 0 146 ohm, maximum

Polarity Additive, with terminals 2 and 3, and 5 and 6 connected

Temperature rise 99°C with 107 volts rms, 400 Hz across (1-2) at an ambient temperature of 71°C

(B) Quality assurance provisions

Qualification inspection Not applicable for this specification sheet

Quality conformance inspection Groups A and B tests of MIL-T-27 shall be applicable

Marking location See figure 1

Part or Identifying Number (PIN) M27/221-01

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0815-24)

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